7.3.G Let a metric space X be covered by subsets $X_i \subset X$ for $i \in \mathbb{Z}$, such that

$$X_i \subset X_{i+1} \subset X_i + \beta_1$$

for all $i \in \mathbb{Z}$ and some fixed $\beta_1 > 0$. Let $P_i\colon X_i + \beta_2 \longrightarrow X_i$ be retractions for $i \in \mathbb{Z}$ and some $\beta_2 > 0$, such that the inequalities

$$|x_1 - x_2| \leqslant 2\alpha \text{ and } \max_{j=1,2} |x_j - P_i(x_j)| \geqslant \beta$$

for some positive α and β imply that $|P_i(x_1) - P_i(x_2)| \leqslant \alpha$ for all x_1 and x_2 in $X_i + \beta_2$ and all $i \in \mathbb{Z}$.

Proposition: *If X is geodesic, if the intersection* $\bigcap_i X_i$ *is hyperbolic (for the metric induced from X) and if*

$$10\beta_1 \leqslant \beta \leqslant \frac{\alpha\beta_2}{10\beta},$$

then X is hyperbolic.

Proof: It suffices to show, according to 7.2.E that X contains no long locally quasigeodesic circle. Now let S^1 be such a circle in X and i_0 be the minimal integer such that $X_{i_0} + \beta_2$ contains S^1. Using the retraction P_{i_0} one can shorten some ark in S^1 of a controlled length (compare 1.7.B) and the proof follows by contradiction. We leave to the reader filling in the details and figuring out the value of the hyperbolicity constant.

Example: If $X = \mathbb{R}^2$ then the pertinent S^1 is the round circle of a sufficiently large radius.

7.3.H Let us give a "local" criterion for ϵ-convexity of a *connected* subset Y in a δ-hyperbolic space X.

Proposition: *Let for a given* $\alpha \geqslant 50\delta$ *the center* z *of every segment* $[y_1,y_2]$ *with* $y_1,y_2 \in Y$ *and* $|y_1-y_2| \leqslant 2\alpha$ *satisfies* $\mathrm{dist}(z,Y) \leqslant \alpha - C\delta$ *for* $C = 50$. *Then* Y *is* ϵ*-convex for* $\epsilon = 20\alpha$.

Proof: Let n be the minimal integer, such that there exists a sequence of points $x_1 = y_1,x_2,x_3,\ldots,x_n = y_2$ in Y with $|x_i-x_{i+1}| \leqslant 2\alpha$ for $i = 1,2,\ldots,n-1$. Since n is minimal no point among x_i satisfies the inequality (1) in 7.1.D. Furthermore, if $|x_i-x_{i+3}| \leqslant 2\alpha + 100\delta$ for some $i = 1,\ldots,n-3$ then the center z on $[x_i,x_{i+3}]$ satisfies

$$\max(|x_i-P_Y(z)|,|P_Y(z)-x_{i+3}|) \leqslant 2\alpha,$$

which also is incompatible with the minimality of n. Hence, the inequalities (2) and (3) cannot hold true for any point among x_i. Then, 7.1.B shows that

$$d_0 = \max_i \mathrm{dist}(x_i,[y_1,y_2]) \leqslant 8\alpha$$

and the proof follows by the triangle inequality.

7.3.I **Remarks:**

(a) Instead of the connectivity of Y one only could require the ϵ-neighborhood $Y + \epsilon \subset X$ to be connected for some $\epsilon < \alpha$.

(b) The inequality $\mathrm{dist}(z,Y) \leqslant \alpha - C\delta$ can be replaced by the following more natural condition

(*) there exists a point $z' \in Y$, such that

$$\max(|y_1-z'|,|z'-y_2|) \leqslant 2\alpha - 2C\delta.$$

(c) One can replace (*) by yet another condition for a given integer $n \geqslant 3$.

$(*)_n$ there exists points $z_1' = y_1, z_2', \ldots, z_n' = y_2$ such that

$$\max_i |z_i' - z_{i+1}'| \leqslant 2\alpha - 2C_n\delta$$

for $C_n \leqslant 50(n-2)$.

The proof of these remarks is left to the reader.

7.3.J Call a metric space X *quasigeodesic* if it admits an isometric embedding into a geodesic space, say $X \hookrightarrow \bar{X}$, such that

$$\operatorname{dist}(\bar{x}, X) \leqslant \epsilon < \infty$$

for all $\bar{x} \in \bar{X}$ and some $\epsilon > 0$. (If one wants to specify ϵ one says X is ϵ-geodesic.)

Using 6.4 and 7.3.H one obtains the following

Proposition: *Let X be a hyperbolic metric space and fix some numbers $\mu > 1$ and an integer $n \geqslant 3$. If for every two points x and y in X there exists points $x_1 = x, x_2, \ldots, x_n = y$ in X, such that*

$$\max_i |x_i - x_{i+1}| \leqslant \mu^{-1}|x-y| + n,$$

then X is quasigeodesic.

7.3.K **Convex hulls:** Take a subset Y in a geodesic space X and let $\bar{Y} = \operatorname{Conv} Y \subset X$ be the union of geodesic segments in X.

Lemma: *If X is δ-hyperbolic then $\bar{Y}$ is ϵ-convex for $\epsilon = 4\delta$.*

Proof: One obviously reduces the lemma to the case of a four point subset Y in X where the proof follows from 6.3.

7.4 <u>**Convexity of the distance functions**</u>: If X is a tree then the distance function $(x,y) \longrightarrow |x-y|$ is convex. Namely, for any two geodesic segments $[x_1,x_2]$ and $[y_1,y_2]$ the distance function is convex on $[x_1,x_2] \times [y_1,y_2]$. Since the definition of convexity uses at most six points in X at a time (namely, x_1, x_2, y_1, y_2, $z_1 = \alpha x_1 + (1-\alpha)x_2$ and $z_2 = \alpha y_1 + (1-\alpha)y_2$ for $0 \leqslant \alpha \leqslant 1$), the approximation lemma in 6.1 shows the distance function to be *ϵ-convex* that is $|z_1-z_2| - \epsilon \leqslant \alpha|x_1-y_1| + (1-\alpha)|x_2-y_2|$ in every δ-hyperbolic space X for $\epsilon \leqslant 6\delta$.

<u>**Remark**</u>: Notice that the function $x \longmapsto |x-x_0|$ for a fixed $x_0 \in X$ is more than just ϵ-convex. In fact, the restriction of this function to every segment $[x_1,x_2] \subset X$ is ϵ-close to the function $\mu + |x-x_\mu|_{\mathbb{R}}$ where $\mu = \inf\limits_{x \in [x_1,x_2]} |x-x_0|_X$ and $x_\mu \in [x_1,x_2]$ is a point where $|x_\mu - x_0|_X = \mu$. Similarly, the distance to an ϵ-convex subset $X_0 \subset X$ is shaped the same way in every segment $[x_1,x_2]$ which does not meet the ϵ-neighborhood of X_0 for $\epsilon = 6\delta$.

7.4.A <u>**The displacement of an isometry**</u>: Consider a map $\gamma: X \longrightarrow X$ and let $d_\gamma(x) = |x-\gamma(x)|$. If γ is an isometry and X is geodesic δ-hyperbolic then the (displacement) function d_γ is ϵ-convex for $\epsilon = 6\delta$ by 7.3.A. Let for an isometry $\gamma: X \longrightarrow X$,

$$\bar{d}_\gamma(\leqslant t) = \operatorname{Conv} d_\gamma^{-1}[0,t] \subset X$$

and observe that

$$\bar{d}_\gamma(\leqslant t) \subset d_\gamma^{-1}[0,t+\epsilon]$$

for all $t \in \mathbb{R}_+$, by the ϵ-convexity of d_γ. Also observe that the subsets $\bar{d}_\gamma(\leqslant t)$ are γ-invariant and 4δ-convex.

Another useful construction of invariant ϵ-convex subsets in

X is as follows. Let Γ be an arbitrary group isometrically acting on X. Then the convex hull of every orbit of Γ clearly is Γ-invariant and 4δ-convex.

Consider a geodesic segment $S = [x_1,x_2]$ in X and apply the above to the function $d^S(y) = \mathrm{dist}(\gamma(y),[x_1,x_2]$ where $y \in [x_1,x_2]$ and $\gamma: X \longrightarrow X$ is an isometry. This yields

$$(*) \qquad d^S(y) \leqslant 6\delta + \max(0, d_\gamma(x_1) - |x_1 - y|, d_\gamma(x_2) - |x_2 - y|).$$

Next, consider a geodesic triangle Δ with vertices $x_i \in X$, $i = 1,2,3$ and edges $[x_i,x_j]$. Since Δ is 2δ-thin, there exists a point $z \in X$ which is 2δ-close to each edge of Δ. Apply (*) to an isometry γ of X and bound the displacement of z by

$$(**) \qquad d_\gamma(z) \leqslant 12\delta + \max_{i=1,2,3}(0, d_\gamma(x_i) - |x_i - z|).$$

It follows that an arbitrary point $z' \in X$ for which $(x_i,x_j)_{z'} \leqslant \epsilon$ for some $\epsilon \geqslant 0$ satisfies

$$(***) \qquad d_\gamma(z') \leqslant 4\epsilon + 16\delta + \max_i(0, d_\gamma(x_i) - |x_i - z'|).$$

7.4.B __Solution of the conjugacy problem__: Consider a group Γ with a word metric and suppose the word problem is solvable in Γ. Then the following condition obviously implies the solution of the conjugacy problem.

There exists a constant $C > 0$ such that for every two conjugate elements γ and γ' there exists $\gamma_1 = \gamma, \gamma_2, \gamma_3, \ldots, \gamma_n = \gamma'$ in Γ such that $|\gamma_i| \leqslant C(|\gamma| + |\gamma'|)$ and $\gamma_{i+1} = \alpha_i \gamma_i \alpha_i^{-1}$ for some $\alpha_i \in \Gamma$ with $|\alpha_i| \leqslant C$ for all $i = 1,\ldots,n-1$. This condition says in the geometric language, that every two homotopic closed curves γ and γ' in a compact manifold V with $\pi_1(V) = \Gamma$ can be joined by a homotopy of closed curves γ_t with length $\gamma_t \leqslant C(\text{length } \gamma + \text{length } \gamma')$. If V has $K \leqslant 0$ such a homotopy is constructed by shortening curves to a closed geodesic. This is essentially equivalent to minimizing the displacement functions of γ and γ' on the universal covering X of V, and then, obviously, the quasiconvexity of the displacements solves the conjugacy problem. In particular we obtain the solution for word hyperbolic groups Γ as well as for discrete cocompact isometry groups of convex spaces. Moreover, in the word hyperbolic case we also have the linear isoperimetric inequality for the conjugation. Algebraically, this means the bound

$$n \leqslant \text{const}(|\gamma| + |\gamma'|).$$

Geometrically, it provides a homotopy A: $S^1 \times [0,1] \longrightarrow V$ between closed curves γ and γ', such that Area A $\leqslant$ const(length γ + length γ'), provided γ and γ' are homotopic in V to start with. Thus linear isoperimetric inequality for disks (which implies hyperbolicity by 6.8) yields such inequality for annuli. As we shall see in 8.3 there is a similar inequality for surfaces in the word hyperbolic case. A similar result (including the solution of the conjugacy problem) is known (and easy to prove) for many classes of semihyperbolic groups, such as small cancellation groups and cocompact isometry groups of (convex) manifolds and spaces with $K \leqslant 0$.

7.5 **Rays, lines, distance-like functions and horofunctions:** A *ray* in a metric space X is (the image of) an isometric map r: $\mathbb{R}_+ \longrightarrow X$. A *line* is (the image of) an isometric map $\mathbb{R} \longrightarrow X$. If X is hyperbolic then there obviously exists a limit

$$r(\infty) \overset{\text{def}}{=} \lim_{t\to\infty} r(t) \in \partial X,$$

called the *end* of the ray r. Similarly, every line ℓ has two ends $\ell(-\infty)$ and $\ell(+\infty)$ in ∂X. We sometimes say that the ray r *joins* r(0) and r(∞) and that the line ℓ joins $\ell(-\infty)$ and $\ell(\infty)$.

Usually rays and lines appear as limits of geodesic segments in X. For example let s_i: $[0,t_i] \longrightarrow X$ be a sequence of isometric maps (segments) with $s_i(0) = x_0$ and $s_i(t) = x_i \longrightarrow a \in \partial X$ for $i \longrightarrow \infty$. If the space X is *proper*, that is if every closed ball in X is compact (e.g. X is a complete Riemannian manifold) then some subsequence of maps s_i converges for every fixed $t_0 \in [0,\infty)$ and the limit map $[0,\infty) \longrightarrow \mathbb{R}$ is a ray in X joining x_0 and a. Thus, in a proper geodesic hyperbolic space X the boundary ∂X equals the set of *asymptotic* classes of rays, where two rays r_1 and r_2 are called *asymptotic* if $|r_1(t)-r_2(t)| \leqslant \text{const} < \infty$ for all $t \in [0,\infty)$.

Now, to construct a line joining a and $b \neq a$ in ∂X we take sequences of points in X say $x_i \longrightarrow a$ and $y_i \longrightarrow b$ such that $\big| |x_i| - |y_i| \big| \leqslant \text{const} < \infty$ for $i \longrightarrow \infty$. We parametrize geodesic segments $[x_i,y_i]$ by isometric maps s_i: $[-t_i,t_i] \longrightarrow X$ (for $2t_i = |x_i-y_i|$) and observe that in the hyperbolic case the centers $s_i(0) \in X$ lie in a fixed (bounded!) ball as $i \longrightarrow \infty$. Therefore some subsequence converges to a line between a and b, provided X is proper.

If X is non-proper, then the above (limit) rays and lines do not always exist. One may use instead quasigeodesic lines and rays which work equally well in many cases. Another possibility is to slightly perturb (or complete) X in order to have actual lines and rays. Namely call a geodesic hyperbolic space Y *ultra-complete* if any two points in $Y \cup \partial Y$ can be joined by a segment, ray or a line in Y. Then, by a trivial argument, every geodesic hyperbolic X isometrically embeds into some ultra-complete space Y, such that $\text{dist}_Y(y,X) \leqslant \text{const} < \infty$ for all $y \in Y$. In particular, $\partial Y = \partial X$. Furthermore, one can choose a Y, such that every isometry of X extends to a unique isometry of Y.

Thus, in most cases, one can work with X as if it were ultra-complete to start with.

7.5.A __Quasiconvex hulls of subsets in the boundary__: If X is a geodesic hyperbolic space then there is an important correspondence between closed subsets $A \subset \partial X$ containing at least two points and certain quasiconvex subsets in X. To go from subsets in X to ∂X one assigns to each $X_0 \subset X$ its boundary $\partial X_0 \subset \partial X$. A more interesting operation is assigning to A the union of all lines in X between the pairs of points in A which is denoted $L(A) \subset X$. If X is ultracomplete and δ-hyperbolic, the, by the arguments in 7.3, this "hull" $L(A)$ is 6δ-convex and $\partial L(A) = A$. If X is not ultracomplete, one first takes $L(A)$ in an ultracompletion $Y \supset X$ (with $\partial Y = \partial X$) and then defines $L'(A) \subset X$ as the intersection of $X \subset Y$ with some ϵ-neighborhood (for possibly large ϵ) of $L(A)$ in Y. An alternative is to define $L(A)$ with quasigeodesics in X.

7.5.B <u>Distance-like functions and forms on X</u>: Our objective now is to describe the boundary ∂X using functions $f\colon X \longrightarrow \mathbb{R}$ rather than rays $\mathbb{R}_+ \longrightarrow X$. The idea comes from the following

7.5.C <u>Example</u>: Start with a ray $r\colon \mathbb{R}_+ \longrightarrow X$ and associate to it the following *ray* (or *Busemann*) *function*

$$f(x) = \lim_{t\to\infty}(|x-r(t)|-t).$$

Observe several obvious properties of this function $f\colon X \longrightarrow \mathbb{R}$.

(1) Every level set $f^{-1}(-\infty,t) \subset X$ is the union of an increasing family of balls in X.

(2) The function f is "distance-like":

$$f(x) = t + \operatorname{dist}(x, f^{-1}(-\infty,t))$$

for all $t \in \mathbb{R}$ and those $x \in X$, where $f(x) \geqslant t$.

(3) If X is hyperbolic and $f(x_i) \longrightarrow -\infty$ for some $x_i \in X$, then $x_i \longrightarrow r(\infty) \in \partial X$ for $i \longrightarrow \infty$.

(4) Take a point $x_1 \in X$ and a sequence of numbers $t_1 > t_2 > \dots > t_i > \dots$ such that $t_1 < f(x_1)$ and $t_i \longrightarrow -\infty$ for $i \longrightarrow \infty$. Define by induction a sequence of points $x_i \in X$ where x_{i+1} is the normal projection of x_i to the level $f^{-1}(-\infty,t_i] \subset X$. Such an x_{i+1} does exist if X is proper. Otherwise we take an ϵ_{i+1}-approximation to such a projection for some sequence $\epsilon_i \longrightarrow 0$

for $i \longrightarrow \infty$. Then we join each x_i with x_{i+1} by a segment $[x_i, x_{i+1}]$ and thus obtain a map, called *a gradient ray* $r_1 \colon \mathbb{R}_+ \longrightarrow X$ issuing from x_1. If the implied constants ϵ_i are all $= 0$, then this is indeed a ray. Otherwise this is quasiray. If X is hyperbolic, then, obviously, this quasiray is asymptotic to the original ray r issuing from x_0.

7.5.D __Horofunctions:__ The above construction of a ray function can be generalized as follows. Fix a reference point $x_0 \in X$ and let a sequence $x_i \in X$, $i = 1, \ldots$, have $|x_i| \longrightarrow \infty$ for $i \longrightarrow \infty$. Consider the sequence of functions on X

$$f_i(x) = |x - x_i| - |x_0 - x_i| .$$

If X is proper, then there is a subsequence which converges on every bounded subset in X. Limits f of this kind are called *horofunctions* on X. If X is geodesic, then horofunctions are very similar to (yet more general than) ray functions.

Now let us see what happens to f_i if X is δ-hyperbolic and $x_i \longrightarrow a \in \partial X$. It is clear that

$$\limsup_{i,\, j \to \infty} |f_i(x) - f_j(x)| \leqslant 4\delta$$

for all $x \in X$. Therefore, even if (no subsequence of) f_i converges, one may take an approximate limit called *quasihorofunction* $f \colon X \longrightarrow \mathbb{R}$, such that $\limsup_{i \to \infty} |f(x) - f_i(x)| \leqslant 4\delta$ for all $x \in X$. These functions satisfy obvious "quasi-fications" of the properties (1)-(4) of ray functions.

Now, let f and f' be two such quasihorofunctions with reference points x_0 and x_0' in X and limit points a and a' in ∂X correspondingly. If $a = a'$, the difference $f - f'$ obviously is 8δ-constant on X,

$$|(f-f')(x) - (f-f')(y)| \leqslant 8\delta$$

for all x and y in X. On the contrary, if $a \neq a'$, then $f - f'$ is very far from a constant. Namely, if X is geodesic, then for every $d > 0$ there exist points x and y in X with $|x-y| = d$, such that

$$|(f-f')(x)-(f-f')(y)| \geqslant 2d - 16\delta.$$

This is seen by looking at f and f' on some (geodesic or quasigeodesic) line $\ell(t)$ in X between a and a'. This line is (quasi)-gradient for both f and f', such that f roughly equals t on $\ell(t)$ and f' is $\approx -t$. Thus $f-f'$ looks like 2t on this line.

7.5.E Let us give a *local* description of horofunctions on a geodesic δ-hyperbolic space X by looking at "differentials" $\varphi(x,y) = f(x)-f(y)$. Namely, consider functions φ such that

(a) $\varphi = \varphi(x,y)$ is defined for all x and y in X satisfying $|x-y| \leqslant 4d$ for some fixed $d \geqslant 100(\delta+1)$.

(b) *Chain (or 1-form) property.*

$$\varphi(x,y) = -\varphi(y,x).$$

(c) *Local exactness.*

$$\varphi(x,y) = \varphi(x,z) + \varphi(z,y),$$

for all x, y and z in X where $\varphi(x,y)$, $\varphi(x,z)$ and $\varphi(z,y)$ are defined.

(d) *Distance-like property.* Fix an $x \in X$ and consider the function $\bar{\varphi}(y) = -\varphi(x,y)$ on the 4d-ball $B_x \subset X$ around x. Then

$$\bar{\varphi}(y) = t + \operatorname{dist}(y, \bar{\rho}^{-1}(-\infty,t))$$

for all t in the interval $-d \leqslant t \leqslant d$ and for all $y \in B_x$, such that $|x-y| \leqslant d$, and $\bar{\varphi}(y) \geqslant t$ and for all $x \in X$.

(e) *Quasiconvexity.* The function $\bar{\varphi}$ is quasiconvex for all

$x \in X$. Namely, if $y = (1/2)(y_1+y_2)$ where $|x-y_1| \leq 2d$ and $|x-y_2| \leq 2d$ then $\bar\varphi(y) \leq (1/2)(\bar\varphi(y_1)+\bar\varphi(y_2)) + 10\delta$.

Remarks: Every φ "integrates" to a function $f(x)$ such that $\varphi(x,y) = f(x) - f(y)$, where f is unique up to an additive constant. This follows from (a), (b) and the simply connectedness of the 2-polyhedron $P_d^2(X)$ (see 1.7).

To avoid too much "quasi"-terminology let us assume that X is proper. Then, starting from any point $x_1 \in X$ we construct a (gradient) sequence $x_1,x_2,\ldots,x_i,\ldots$, such that $|x_i-x_{i+1}| = d$ for all $i = 1,2,\ldots$, and

$$\varphi(x_i,x_{i+1}) = d, \qquad i = 1,2,\ldots$$

as follows. For every $i = 1,2,\ldots$, take $\bar\varphi_i(y) = -\varphi(x_i,y)$ and take for x_{i+1} the normal projection of x_i to the level $\bar\varphi_i^{-1}(-\infty,d)$. It follows from the previous discussion that x_i converge to some point $a = a(\varphi) \in \partial X$ independent of the starting point $x_1 \in X$. Next we observe as earlier the following implications for every two φ_1 and φ_2 satisfying (a)-(e).

$$|\varphi_1-\varphi_2|_d \leq 20\delta \Rightarrow a(\varphi_1) = a(\varphi_2) \Rightarrow |\varphi_1-\varphi_2|_d \leq 10\delta,$$

where

$$|\varphi_1-\varphi_2|_d = \sup_{|x-y|\leq d} |\varphi_1(x,y)-\varphi_2(x,y)|.$$

Therefore, the inequality $|\varphi_1-\varphi_2|_d \leq 20\delta$ is an equivalence relation on our functions and ∂X is identified with the set of equivalence classes. Notice that both the definition of the set Φ of functions φ (by (a)-(e)) and the equivalence relation on this set are local: in order to decide whether some φ lies in our set and whether $|\varphi_1-\varphi_2|_d \leq 20\delta$ one only should look at φ (and φ_1 and φ_2) in the 4d-ball around each point in X. Also observe that $(\Phi, |\ |_d \leq 20\delta)$ is invariant under all isometries of X. Finally assume the balls $B_x \subset X$ of radius 4d around all points $x \in X$ to be *uniformly compact.* That is each B_x can be covered by N balls of radius ϵ for all $\epsilon > 0$ where N depends only on ϵ.

7.5.F <u>**Proposition**</u>: *The quotient map*

$$\pi: \Phi \longrightarrow \partial X = \Phi / |\ |_d \leq 20\delta$$

is compact-to-one for the norm $|\ |_d$ *in* Φ.

<u>**Proof**</u>: Take a point $a \in \partial X$ and a sequence of points $x_i \longrightarrow \partial X$ on some ray in X converging to a. If two functions φ and φ' in Φ with $a(\varphi) = a(\varphi') = a$ are equal on all balls B_{x_i} then $\varphi \equiv \varphi'$. This follows from the contracting property of the normal projection to quasi-convex sets (see 7.3) and the distance-like property of the (integrated) functions φ. Moreover, if $|\varphi - \varphi'|_d$ is bounded by some ϵ on infinitely many balls B_{x_i} then $|\varphi-\varphi'|_d \leq \epsilon$. Hence the pull-back $\pi^{-1}(a) \subset \Phi$ is compact for all $a \in \partial X$. Q.E.D.

7.5.G Let us specialize the above discussion to a locally finite 1-dimensional complex X with a simplical metric having the edges of length one. Denote by $\Phi_0 \subset \Phi$ the subset of function $\varphi(x,y)$ taking *integral* values on the vertices x and y of the complex X. Notice that Φ_0 naturally is the projective limit of finite sets and thus Φ_0 is homeomorphic to a Cantor set. The map $\Phi \longrightarrow \partial X$ restricts to a *surjective* map $\pi_0: \Phi_0 \longrightarrow \partial X$ which by the above argument is *finite-two-one*. In fact it is clear that the number $\#\pi_0^{-1}(a)$ does not exceed $(8d)^N$ where N is an upper bound to the number of vertices in X lying in a ball $B_x \subset X$ of radius 4d for all $x \in X$.

7.5.H The above "coding" of ∂X by Φ (and by Φ_0) is invariant under all isometries of X. Let us give a quasi-isometry invariant "coding" of ∂X by "quasigradient fields" on X rather than by "forms" on X.

A *field* ψ on X by definition is a subset $\psi \subset X \times X$ whose two projection to X are onto and such that

$$\text{length}\ \psi \underset{\text{def}}{=} \sup_{(x,y)\in\psi} |x-y| < \infty.$$

For example, with every function f on X one may associate

$$\psi = \text{grad}_{r,d}\ f = \{x,y \mid |x-y| \leqslant d,\ f(x)-f(y) \geqslant \lambda\}.$$

An *orbit* of a field ψ is a finite or infinite sequence of points x_i in X, such that $(x_i,x_{i+1}) \in \psi$ for all i. A field ψ of finite length is called *quasigeodesic* if every orbit $x_1,x_2,\dots,x_k$ has $|x_1-x_k| \geqslant \lambda k$ for some number $\lambda = \lambda(\psi) > 0$. Call ψ *non-expanding* if every two orbits $x_1,\dots,x_k$ and $y_1,\dots,y_\ell$ with $|x_1-y_1| \leqslant \epsilon$ and $\big| |x_1-x_k|-|y_1-y_\ell| \big| \leqslant 2$ length ψ satisfy

$$|x_k-y_\ell| \leqslant C + \mu(|x_1-x_k|+|y_1-y_\ell|),$$

where $\epsilon > 0$, $C \geqslant 0$ and $\mu < 1$ are some constant depending on ψ.

Now, if ψ is quasigeodesic then every infinite orbit of ψ converges to some point $a \in \partial X$ and if ψ is non-expanding this points a only depends on ψ (but not on a choice of an orbit of ψ). This gives a surjective map $\beta: \Psi \longrightarrow \partial X$, where Ψ is the space of the quasigeodesic non-expanding fields ψ on X. Furthermore, observe that $\beta(\psi_1) = \beta(\psi_2) \Longleftrightarrow \psi_1 \vee \psi_2 \in \Psi$. Thus we obtained a quasi-isometry invariant "coding" of ∂X by fields $\psi \in \Psi$. This "coding: is *local* in the following sense. Given numbers ℓ, λ, ϵ, C and μ. Then for a field ψ of length $\psi \leqslant \ell$ the above conditions with these constants only need checking within each ball of radius R in X for some R depending on ℓ, λ, ϵ, C and μ. Furthermore, the equivalence relation $\beta(\psi_1) = \beta(\psi_2)$ between fields also is local. It is also clear that the set Ψ and the equivalence relation are quasi-isometry invariant.

7.6 <u>Coding ∂X with trees</u>: Fix a point $x_0 \in X$ and consider the spheres $S_i \subset X$ for $i = 0,1,2,\ldots,$

$$S_i = \{x \in X \mid |x_i| = i\}.$$

Take a subset $V \subset S_i$ for $i \geqslant 1$ and let $P(V) \subset S_{i-1}$ be the normal projection of V to S_{i-1} defined by

$$P(V) = \{x \in S_{i-1} \mid \operatorname{dist}(x,V) = 1\}.$$

Define for a subset $V \subset S_i$

$$\operatorname{Diam}_k V = \sup_j \operatorname{Diam} P^j(V),$$

where j runs over $0,1,2,\ldots,\max(i,k)$ and where $P^j = \underbrace{P \circ P \circ \cdots \circ P}_{j}$.

Now let X be geodesic δ-hyperbolic and consider the graph(1-complex) $T(X)$ whose vertices correspond to the subsets $V \subset S_i$ with $\operatorname{Diam}_{100} V \leqslant 10\delta$ for all $i = 0,1,\ldots,$ and where V_1 and V_2 are joined by an edge if and only if $V_2 = P(V_1)$. Clearly, this $T = T(X)$ is a tree growing from x_0. Denote by ∂T the hyperbolic boundary of T for the simplical metric with unit edges and observe that every geodesic ray in T issuing from x_0 is represented by a family of asymptotic rays in X. Thus we obtain a continuous map of ∂T onto ∂X.

Let us specialize to the case where X is itself a graph with a simplical metric having unit edges, such that each vertex in X has at most $N < \infty$ adjacent edges. Then the tree T also is locally finite with at most $\exp\exp(\delta+1)N$ edges at every vertex and the map $\partial T \longrightarrow \partial X$ is finite-to-one where "finite" $\leqslant \exp\exp\exp(\delta+1)N$. Checking all this is left to the reader.

§8. Isometry groups of hyperbolic spaces

According to 6.4 every hyperbolic metric group isometrically acts on some *geodesic* hyperbolic space - and by 7.5 this X can be assumed *ultracomplete*. In this section we fix such a ultracomplete δ-hyperbolic space X and study groups Γ isometrically acting on X.

8.1 Classification of individual isometries: An isometry $\gamma: X \longrightarrow X$ is called *elliptic* if the orbit $\{\gamma^i x\} \subset X$, $i \in \mathbb{Z}$, is bounded for some (and hence, for every) $x \in X$. Call γ *parabolic* if the orbit $\{\gamma^i x\}$ has a unique *limit point* $a \in \partial X$. That is the orbit is unbounded and $|\gamma^{i_j} x| \longrightarrow \infty$ implies $\gamma^{i_j} x \longrightarrow a$ for every subsequence $\{i_j\} \subset \mathbb{Z}$. A parabolic isometry is called *proper* if $|\gamma^i x| \longrightarrow \infty$ for $|i| \longrightarrow \infty$. Finally, an isometry γ is called *hyperbolic* if the orbit map $\mathbb{Z} \longrightarrow X$ for $i \longmapsto \gamma^i x$ is a quasi-isometry for the ordinary metric in $\mathbb{Z}$. If γ is hyperbolic, then, obviously, the limit set of the orbit $\{\gamma^i x\}$ consists of exactly two points:

$$\gamma^{\infty} = \lim_{i \to +\infty} \gamma^i x \text{ and } \gamma^{-\infty} = \lim_{i \to -\infty} \gamma^i x \text{ in } \partial X.$$

Notice that γ^{∞} and $\gamma^{-\infty}$ are independent of x. Also observe that the above classification of isometries is stable under replacement of γ by γ^{i_0} for any $i_0 \neq 0$.

8.1.A Lemma: *Let* γ_1 *and* γ_2 *be two non-hyperbolic isometries of* X *such that the points* $x_j = \gamma_j(x)$ *for* $j = 1,2$, *satisfy*

$$(*) \qquad |x - x_j| \geqslant 2(x_1.x_2)_x + 400\delta.$$

Then the isometries $\gamma_1\gamma_2$ *and* $\gamma_2\gamma_1$ *are hyperbolic.*

Proof: Since γ_1 and γ_2 are non-hyperbolic, Lemma 7.2.C (applied to the sequences $\gamma_1^i x$ and $\gamma_2^i x)$ shows that $|\gamma_j^2(x)| \leqslant |\gamma_j(x)| + 100\delta$ for $j = 1,2$. Then, by (*) and the δ-hyperbolicity of X, the isometries $\gamma_1\gamma_2$ and $\gamma_2\gamma_1$ satisfy the opposite inequality which yields via 7.2.C the hyperbolicity of these isometries.

8.1.B **Corollary**: *Every isometry γ is elliptic, parabolic or hyperbolic.*

Proof: If γ is non-elliptic and non-parabolic, then there exists two integers i_1 and i_2, such that the Lemma applies to γ^{i_1} and γ^{i_2}.

8.1.C Let us look at the displacement function $d_\gamma(x)$ of γ (see 7.4). If γ is hyperbolic then $d_\gamma(x)$ is bounded on the geodesic $\ell \subset X$ joining the two limit points and $d_\gamma(x) \longrightarrow \infty$ for $\mathrm{dist}(x,\ell) \longrightarrow \infty$ as is seen by looking at the normal projection of x to ℓ. If γ is parabolic, then the $\delta_\gamma(x)$ is bounded on the ray r in X between x and the limit $a \in \partial X$. The translates $\gamma^i(r)$ are 4δ-close to r near infinity for all $i \in \mathbb{Z}$. It follows that for every $i_0 = 1,2,$ there exists a point $x_0 \in X$ such that $d_{\gamma^i}(x_0) \leqslant 8\delta$ for $|i| \leqslant i_0$.

8.1.D **Corollary**: *If Γ is a word hyperbolic group, then every non-torsion element γ in Γ is hyperbolic. That is the homomorphism $\mathbb{Z} \longrightarrow \Gamma$ sending $i \mapsto \gamma^i$ is quasi-isometric.*

This corollary shows that if two elements γ_1 and γ_2 in Γ are conjugate, then there exists an $\alpha \in \Gamma$, such that $\alpha\gamma_1\alpha^{-1} = \gamma_2$ and $|\alpha| \leqslant 2(|\gamma_1| + |\gamma_2| + 8\delta)$. Since the norm $\gamma \longmapsto |\gamma|$ is a primitive recursive function (by 2.3), we see that *the conjugacy problem in Γ is solvable* (compare 7.4).

8.1.E The geometric properties of isometries (see 7.4) and quasi-convex sets in X (see 7.3) lead to the following dynamical corollaries as closed subsets A in ∂X generate (or span) quasi-convex subsets $L(A)$ in X (see 7.5).

8.1.F *Let γ be proper parabolic and let $U \subset X \cup \partial X$ be an arbitrary neighborhood of the limit point $a \in \partial X$. Then there exists an integer i_0 such that the complement $(X \cup \partial X)\setminus U$ is sent to U by the isometry γ^i for all $i \geqslant i_0$.*

8.1.G *Let γ be hyperbolic and let U_+ and U_- in $X \cup \partial X$ be some neighborhoods of the limit points $\gamma^{+\infty}$ and $\gamma^{-\infty}$ in ∂X correspondingly. Then there is an i_0, such that $(X \cup \partial X)\setminus U_-$ is sent by γ^i to U_+ for all $i \geqslant i_0$ and $(X \cup \partial X)\setminus U_+$ is sent by γ^{-i} to U_-. Furthermore if i_0 is sufficiently large then the maps*

$$\gamma^i\colon \partial X\setminus U_- \longrightarrow \partial X \cap U_+$$

and

$$\gamma^{-i}\colon \partial X\setminus U_+ \longrightarrow \partial X \cap U_-$$

are 2-contracting for the metric in ∂X introduced in 7.2.

(A map $f\colon A \longrightarrow B$ is called 2-contracting if $|f(a_1)-f(a_2)| \leqslant 1/2|a_1-a_2|$ for all a_1 and a_2 in A.)

8.2 Non-elementary isometry groups Γ and their action on $\partial\Gamma$, $\partial^2\Gamma$ and $\partial^3\Gamma$: Let Γ be a non-elliptic isometry group of Γ, that is some (and hence every) orbit $\{\Gamma x\} \subset X$ for $x \in X$ is unbounded. Then we see as in 8.1 that either Γ is *parabolic*, that is there is a unique limit point $a \in \partial X$ or Γ contains a hyperbolic element γ. Recall that the limit set $\partial\Gamma \subset \partial X$ by definition is the hyperbolic boundary of an orbit $\{\Gamma x\} \subset X$ which clearly is independent of $x \in X$.

8.2.A **Lemma**: *Let* $A \subset \partial X$ *be a closed* Γ*-invariant subset containing at least two points. Then* $A \supset \partial\Gamma$.

Proof: Take $L(A) \subset X$ (see 7.5) and $x \in L(A)$. Then $\{\Gamma x\} \subset L(A)$ and the lemma follows.

8.2.B If $\partial\Gamma = \partial\{\gamma^i\}_{i\in\mathbb{Z}}$ for some hyperbolic element $\gamma \in \Gamma$, then by the proof of 8.2.A *each orbit* $\{\Gamma x\} \subset X$ *is quasi-isometric to* $\mathbb{Z}$.

8.2.C **Corollary**: *Every non-torsion element* γ *in a word hyperbolic group* Γ *is contained in a unique maximal elementary subgroup* $E_\gamma \subset \Gamma$ *which is a finite extension of the infinite cyclic group generated by* γ. *Hence* $\gamma = \gamma_0^p$ *for some prime element* $\gamma_0 \in \Gamma$ *and* $p \in \mathbb{N}$. *The exponent* $p = p(\gamma)$ *is unique. Moreover if* Γ *has no torsion then* $\gamma_0 = \gamma_0(\gamma)$ *also is unique.*

8.2.D Now, let $\partial\Gamma$ contain at least three points. Then there exists at most one point $a \in \partial\Gamma$ fixed under Γ. If such a exists then Γ is called *quasiparabolic*. It follows from 3.2.A that Γ acts *minimally* away from a. That is the orbit of every non-fixed point $b \in \partial\Gamma$ is dense in $\partial\Gamma$. Hence, the points $\gamma^\infty \in \partial X$ for the hyperbolic $\gamma \in \Gamma$ are dense in ∂X and therefore $\partial\Gamma$ has no isolated points. Since $\partial\Gamma$ is complete it is uncountable. In this case Γ is called *non-elementary*.

8.2.E If Γ is non-elementary then, by the above, there are two hyperbolic elements γ_1 and γ_2 in Γ, such that $\gamma_2^{+\infty} \neq \gamma_1^{\pm\infty}$. Take small neighborhoods $U_j \subset \partial\Gamma$ of $\gamma_j^{+\infty}$ for $j = 1,2$ and observe with that

$$\gamma_j^i(U_1 \cup U_2) \subset U_j$$

for $j = 1,2$, for all $i \geqslant i_0$ and for some $i_0 > 0$. It follows that the semigroup F_2^+ generated by $\gamma_1^{i_1}$ and $\gamma_2^{i_2}$ in Γ is *free* provided i_1 and i_2 are $\geqslant i_0$. Furthermore there is a unique closed subset in $U_1 \cup U_2$ univariant under F_2^+ which is homeomorphic to the Cantor set.

8.2.F Now let Γ be non-quasiparabolic as well as non-elementary. Then, by an obvious argument, for any two non-empty open subsets U_1 and U_2 in $\partial\Gamma$ there exists hyperbolic elements γ_1 and γ_2, such that $\gamma_1^{+\infty} \in U_1$, $\gamma_2^{+\infty} \in U_2$ and $\gamma_1^{-\infty} \neq \gamma_2^{-\infty}$. Take disjoint neighborhoods U_1^- and U_2^- in $\partial\Gamma$ of $\gamma_1^{-\infty}$ and $\gamma_2^{-\infty}$ correspondingly and assume that U_1 and U_2 also are disjoint. Take a sufficiently large i_0, such that γ_1^i sends $U_1 \cup U_2 \cup U_2^-$ to U_1 for $i \geqslant i_0$ (see 8.1) and γ_1^{-i} sends $U_1^- \cup U_2 \cup U_2^-$ to U_1^-, while the transformations γ_2 send $U_2 \cup U_1 \cup U_1^-$ and $U_2^- \cup U_1 \cup U_1^-$ to the subsets U_2 and U_2^- correspondingly. It immediately follows that the subgroup $F \subset \Gamma$ generated by $\gamma_1^{i_1}$ and $\gamma_2^{i_2}$ is free if i_1 and i_2 are $\geqslant i_0$, and that all isometries in this subgroup are hyperbolic. Furthermore the orbit map $f \longmapsto fx$ is a quasi-isometry of F into X for the word metric in F. The extension of this quasi-isometry to ∂F (which is homeomorphic to the Cantor set) sends ∂F into the union $U_1 \cup U_2 \cup U_1^- \cup U_2^-$. (For large i_0 this ∂F equals the maximal F-invariant subset in this union.) Finally we observe that the product $\alpha = \gamma_1^{i_1}\gamma_2^{-i_2}$ has $\alpha^{+\infty} \in U_1$ and $\alpha^{-\infty} \in U_2$.

8.2.G <u>Corollary</u>: *The set of pairs* $(\gamma^{+\infty},\gamma^{-\infty}) \in \partial\Gamma \times \partial\Gamma$ *for all hyperbolic* $\gamma \in \Gamma$ *is dense in* $\partial\Gamma \times \partial\Gamma$. *In particular, every non-elementary word hyperbolic group* Γ *has infinitely many conjugacy classes of prime non-torsion elements.*

By a similar argument one obtains

8.2.H *The action of* Γ *on* $\partial\Gamma \times \partial\Gamma$ *is topologically transitive. That is every open non-empty* Γ*-invariant subset is dense.*

Since $\partial\Gamma$ is complete one concludes,

8.2.I *There is a point in* $\partial\Gamma \times \partial\Gamma$ *whose* Γ*-orbit is dense.*

8.2.J **Expansion points:** A point $a \in \partial X$ is called an E-*point* for Γ if for every $\lambda \geqslant 1$ there exists a neighborhood $U \subset \partial X$ of a and an element $\gamma \in \Gamma$, such that the action of γ on U is λ-*expanding* for the metric on ∂X defined in 7.2. That is

$$|\gamma(a_1)-\gamma(a_2)| \geqslant \lambda |a_1-a_2|$$

for all a_1 and a_2 in U. It is easy to see that all E-points lie in the limit set $\partial\Gamma \subset \partial X$. In fact, by the definition of the metric in ∂X, a point $a \in \partial X$ is an E-point if and only if the intersection of the Γ-orbit of some ball $B \subset X$ with some ray r in X from a fixed point $x_0 \in X$ to a is unbounded,

$$\mathrm{Diam}(\Gamma B) \cap r = \infty.$$

For example, if the quotient X/Γ is bounded then every $a \in \partial X$ is an E-point. In particular, *if* Γ *is word hyperbolic then every* $a \in \partial\Gamma$ *is an* E-*point*. More generally let Γ' be a subgroup in the word hyperbolic group Γ and let $\partial\Gamma' \subset \partial\Gamma$ be the limit set. *Then the following two conditions are equivalent.*

(1) *All points* $a' \in \partial\Gamma'$ *are* E-*points for* Γ';

(2) Γ' *is word hyperbolic (and, hence finitely*

presented) and the inclusion $\Gamma' \hookrightarrow \Gamma$ *is quasi-isometric for the word metrics in* Γ' *and* Γ.

8.2.K __Action of Γ on $\partial^3 X$:__ Let $\partial^3 X \subset \partial X \times \partial X \times \partial X$ be the set of triples (a_1,a_2,a_3), where $a_1 \neq a_2 \neq a_3 \neq a_1$. To each point $(a_1,a_2,a_3) \in \partial^3 X$ one assigns a *bounded* subset $C \subset X$ as follows

$$C = C_\epsilon = \{x \in X \mid \operatorname{dist}(x,\ell_{ij}) \leqslant \epsilon\},$$

for $i,j = 1,2,3$ and for all lines ℓ_{ij} in X between a_i and a_j. If $\epsilon \geqslant 10\delta$, then, clearly, the set C_ϵ is non-empty. This construction establishes a correspondence between bounded subsets in X and *codiagonal* subsets $A \subset \partial^3 X$, where "codiagonal" is defined by

$$\inf \min(|a_1-a_2|, |a_2-a_3|, |a_1-a_3|) > 0,$$

where inf is taken over all $(a_1,a_2,a_3) \in A$ and the distance is measured in ∂X. For example, if ∂X is compact then "codiagonal" amounts to "relatively compact".

The above "bounded $\Longleftrightarrow$ codiagonal" correspondence leads, in particular, to the following

8.2.L __Proposition–Definition:__ *A set* Δ *of isometries of* X *is called bounded if one of the following four equivalent conditions is satisfied.*

(1) $\sup_{\gamma \in \Delta} |x-\gamma(x)| < \infty$ *for a fixed* $x \in X$.

(2) *The set of homeomorphisms* $\gamma\colon \partial X \longrightarrow \partial X$ *for* $\gamma \in \Delta$ *is uniformly continuous.*

(3) *There exists a codiagonal subset* $A \subset \partial^3 X$, *such that* $A \cap \gamma(A)$ *is non-empty for all* $\gamma \in \Delta$.

(4) *There exists an* $\epsilon = \epsilon(\Delta) > 0$ *such that for every*

ϵ-ball $B \subset \partial X$ *and each* $\gamma \in \Delta$ *the complement of* $\gamma(B)$ *satisfies*

$$\mathrm{Diam}(\partial X \setminus \gamma(B)) \geqslant \epsilon.$$

Notice that (4) implies that Δ is bounded $\Longleftrightarrow$ Δ^{-1} is bounded. A related notion is that of B-*finiteness* of an isometry group Γ of X which is defined by the conditions: every bounded subset Δ in Γ is finite. If X is a Riemannian manifold this amounts to the usual discreteness of Γ. Here is a similar generalization of cocompactness.

8.2.M __Definition__: An action of Γ is called *cobounded* if one of the following two equivalent (by the previous discussion) conditions is satisfied.

(1) There is a bounded subset $Y \subset X$, such that $\Gamma(Y) = X$.

(2) There is a codiagonal subset $A \subset \partial^3 X$, such that $\Gamma(A) = \partial^3 X$.

Notice, that for word hyperbolic groups Γ the boundary $\partial\Gamma$ is compact and the action of Γ on $\partial^3 X$ is discrete cocompact (hence it is B-finite and cobounded).

8.2.N __Example__: If X is a complete convex manifold with $K < -\epsilon < 0$, then the space $\partial^3 X$ can be identified with the total space $S_2 X$ of the orthonormal 2-bundle of X by the following well-known construction of Ahlfors-Cheeger. Take a point $(a_1,a_2,a_3) \in \partial^3 X$ join a_1 and a_2 by the (unique for $K < -\epsilon < 0$) line $(a_1,a_2) \subset X$ and observe that the normal projection $P\colon X \longrightarrow (a_1,a_2)$ continuously extends to $\partial X \setminus \{a_1,a_2\}$. Now we define a homeomorphism $h\colon \partial^3 X \longrightarrow S_2 X$ by sending each (a_1,a_2,a_3) to the frame of vectors at $P(a_3) \in (a_1,a_2) \subset X$ tangent to the line (a_1,a_2) and to the ray $[P(a_3),a_3) \subset X$. Notice that h commutes with isometries and thus "descends" to quotients X/Γ.

8.2.P __Corollary__ (Cheeger): *Let* V_1 *and* V_2 *be* *compact manifolds with* $K < 0$ *and let* Γ_1 *and* Γ_2 *be their fundamental groups. Then every isomorphism* $\Gamma_1 \longrightarrow \Gamma_2$ *induces a homeomorphism of (the total space of) the 2-frame bundle of* V_1 *to that of* V_2, *such that compositions of isomorphisms go to compositions of homeomorphisms.*

__Example:__ The group of exterior automorphisms of the fundamental group of every closed surface V acts by homeomorphisms on the unit tangent bundle of V (which is essentially the same as the 2-frame bundle for dim V = 2).

8.2.Q __Remark:__ Many properties of hyperbolic groups can be seen by looking at the action of Γ on ∂X, without ever mentioning X. The key property to use is the equivalence

$$(*) \qquad (2) \Longleftrightarrow (3) \Longleftrightarrow (4)$$

for all $\Delta \subset \Gamma$ (see 8.2.L). It is conceivable that the existence of a complete metric space D and of an action of a group Γ on D by uniform homeomorphisms satisfying (*) makes Γ hyperbolic for a suitable metric on Γ. For example, let "bounded" stand for "codiagonal" in $D \times D \times D$ and consider a B-finite (now B is for codiagonal rather than for bounded) and cobounded action of a group Γ on D. Such a Γ looks very much like a word hyperbolic one. Notice that B-finite cobounded actions are "rather rigid": for two such actions of Γ on D_1 and on D_2 every Γ-equivariant map $D_1 \longrightarrow D_2$ is injective.

As another example let us generalize 8.1 (compare [GM]).

8.2.R *Let* $\gamma: D \longrightarrow D$ *be a uniformly continuous map such that for some* $k \geqslant 2$ *one of the following three conditions is satisfied.*

(a_k) *For every codiagonal subset* A *in*

$D^k = \underbrace{D \times D \times \cdots \times D}_{k}$ *there is at most finitely many* $i \in \mathbb{Z}_+$, *such that* $A \cap \gamma^i(A) \neq \emptyset$, *where "codiagonal" means* $\mathrm{dist}(A,\Sigma) > 0$ *for the union* Σ *of the diagonals (defined by* $d_i = d_j$ *for* $(d_1,\ldots,d_k) \in D$*) in* D.

(b_k) *For every codiagonal* A *there is an integer* m *such that every orbit of* γ *(acting on* D^k*) meets* A *at most* m*-times. (Here "an orbit" means a map* $\mathbb{Z}_+ \longrightarrow D^k$ *and the meeting points with* A *are the pull-backs of* A*; thus every fixed point in* A *meets* A *infinitely many times).*

(c_k) *For the above* A *there is an* m, *such that* A *contains no segment of any orbit of length* m, *that is* $x, \gamma(x), \gamma^2(x), \ldots, \gamma^m(x)$, *for* $x \in D^k$.

Then there is a non-empty finite γ*-invariant subset* $D_0 \subset D$, *such that every orbit asymptotically approaches* D_0.

The proof is straightforward and we leave it to the reader. Here we only notice that $(a_k) \Rightarrow (b_k) \Rightarrow (c_k)$ and that (a_3) is our earlier B-finiteness condition.

8.2.S The above result extends to many groups and semigroups acting on D. Furthermore, if D is compact and Γ is an *amenable* group acting on D, such that no point in $D^k \setminus \Sigma$ is recurrent (this generalizes (c_k)), then there exists a non-empty finite Γ-invariant subset $D_0 \subset D$. In fact, the non-recurrency of $D^k \setminus \Sigma$ condition shows that the support of every Γ-invariant measure on D is finite.

Remark: Many interesting actions of Γ on D have invariant *non-diagonal* subsets $\Sigma \subset D^k$ to which the above discussion applies. For example, if D is a projective space then a useful Σ consists of k-tuples of points lying in a (k-2)-subspace. A fundamental result here is

Tits' free group theorem: *If a finitely generated group Γ of projective transformations of $\mathbb{C}P^n$ contains no free subgroups then there is a nonempty finite union of* k*-dimensional subspaces in $\mathbb{C}P^n$ invariant under Γ for all* $k = 0,1,\ldots,n-1$.

(See [Ti] and compare [Fu].)

8.3 Geodesic flows and hyperbolic ***simplices*****:** Start with two geodesic rays or segments r_1 and r_2 in a convex manifold X with $K \leq -4\epsilon^2 < 0$ which converge to some point $a \in X \cup \partial X$. Then one knows that r_1 and r_2 converge *exponentially* fast. Namely, one can parametrize r_1 and r_2 by length,

$$r_i\colon [0,t_i] \longrightarrow X \quad i = 1,2 \text{ and } r_i \in [0,\infty],$$

such that

$$(+) \qquad |r_1(t+\theta)-r_2(t+\theta)| \leq 2(\exp-\epsilon\theta)|r_1(t)-r_2(t)|,$$

for all $t \in [0,t_1) \cap [0,t_2)$ and for those $\theta \geq 0$ which satisfy

$$t + \theta \in [0,t_1) \cap [0,t_2)$$

and

$$\theta \geq 2|r_1(t)-r_2(t)|.$$

In particular every two distinct points a and b in $X \cup \partial X$ can be joined by a *unique* geodesic in X. It follows that the space of double-infinite geodesics in X is canonically homeomorphic to $\partial^2 X = \{x,y \in \partial X \mid x \neq y\}$.

If X is a general hyperbolic space then one might have several geodesics joining two points in X or in ∂X. Our goal is to find a quasi-isometric perturbation of X to another space X' which would satisfy some version of (+). This is quite easy to achieve if one does not care about the isometries. But what one needs is an X' isometrically acted upon by the isometry group Γ of X. Here one can meet a non-trivial cohomological obstruction even for an X which is quasi-isometric to $\mathbb{R}$.

8.3.A __Example:__ Let Γ be an infinite group, such that $H^1(\Gamma;\mathbb{R}) = 0$ and which admits an unbounded *quasi-homomorphism* $h: \Gamma \longrightarrow \mathbb{R}$. That is $h(\gamma) = h(\gamma^{-1})$ for all $\gamma \in \Gamma$ and

$$|h(\gamma_1\gamma_2)-h(\gamma_1)-h(\gamma_2)| \leqslant d$$

for some $d > 0$ and all γ_1 and γ_2 in Γ. It follows from [Gr2] that every non-elementary word hyperbolic group Γ (e.g. $\Gamma = \{\alpha,\beta \mid \alpha^7 = 1, \beta^7 = 1\}$ which has $H^1(\Gamma;\mathbb{R}) = 0$) admits such an h.

Now, take a constant c, such that $h^{-1}[0,c] \subset \Gamma$ generates Γ and consider the word metric $|\ |$ in Γ corresponding to the (infinite!) generating subset $h^{-1}[0,c'] \subset \Gamma$ for $c' = 10(c+d)$. Clearly the map $h: (\Gamma,|\ |) \longrightarrow \mathbb{R}$ is quasi-isometric. On the other hand since $H^1(\Gamma;\mathbb{R}) = 0$, there is no (non-trivial) isometric action of Γ on $\mathbb{R}$. Notice that $\mathbb{R}$ is the only candidate for X' in this case and so $(\Gamma,|\ |)$ can not be *equivariantly* "straightened out".

This example may seem non-conclusive, as the action of Γ on $\partial\Gamma = \partial\mathbb{R} = \{\pm\infty\}$ is trivial. One gets a true example by taking a free product of two copies of $(\Gamma,|\ |)$.

Notice that amenable (in particular finite) groups have trivial bounded cohomology (see [Gr2]) and so the above phenomenon can not appear for B-finite (or amenable) Γ-action. In fact we shall see below that every word hyperbolic group does admit a well behaved "geodesic flow".

8.3.B First we define *the geodesic flow* for an arbitrary metric space X by considering the space GX of all isometric maps $g: \mathbb{R} \longrightarrow X$ with the metric

$$|g_1-g_2|_{GX} = \int_{-\infty}^{+\infty} \left[|g_1(t)-g_2(t)|_X \right] 2^{-|t|}\, dt.$$

The translations in $\mathbb{R}$ define a (non-isometric) action of $\mathbb{R}$ on GX called the *geodesic flow*. Observe that the projection $GX \longrightarrow X$ given by $g \longmapsto g(0)$ is a quasi-isometry. Also observe that the isometry group Is = IsX isometrically acts on GX and the actions of Is and of $\mathbb{R}$ agree in an obvious sense. Furthermore, the group $\mathbb{Z}_2$ acts on GX by the involution $g(t) \longleftrightarrow g(-t)$ and this action of $\mathbb{Z}_2$ agrees with those of Is and $\mathbb{R}$.

Now, let X be hyperbolic and ultracomplete. Then there is a surjective projection

$$D\colon GX \longrightarrow \partial^2 X = \{a,b \in \partial X \mid a \neq b\},$$

given by $g \longmapsto (g(-\infty), g(+\infty))$. The pull-back $D^{-1}(a,b)$ consists, in general, of several $\mathbb{R}$-orbits. Our main objective is to construct a quotient space of GX by identifying these orbits to a single one for all $(a,b) \in \partial^2 X$, such that a given action on an isometry group Γ on X (and hence on GX) goes to an action on the quotient space. Our example indicates a cohomological obstruction to such an identification of orbits. In particular, one can not expect any *canonical* construction of the desired quotient space. However we have the following

8.3.C **__Theorem:__** *Let Γ be a word hyperbolic group. Then there exists a locally compact finite dimensional hyperbolic metric space $\hat{G}$ acted upon by the groups Γ, $\mathbb{R}$ and $\mathbb{Z}_2$ such that the following six conditions are satisfied.*

(1) *The three actions agree in the same way as those on* GX.

(2) *The actions of Γ and of $\mathbb{Z}_2$ are isometric.*

Furthermore, the action of Γ *is* B-*finite (i.e. discrete) and the orbit map* $\Gamma \longrightarrow \hat{G}$ *for* $\gamma \longmapsto \gamma g_0$ *is a quasi-isometry for every* $g_0 \in \hat{G}$. *This quasi-isometry continuously extends to a canonical (in particular, independent of* g_0*)* Γ*-equivariant homeomorphism* $\partial\Gamma \leftrightarrow \partial\hat{G}$.

(3) *The action of* $\mathbb{R}$ *on* $\hat{G}$ *is free and every orbit* $\mathbb{R} \longrightarrow \hat{G}$, *denoted by* $t \longmapsto g(t)$, *is a quasi-isometric embedding. The map* $g \longmapsto (g(-\infty), g(+\infty)) \in \partial^2\hat{G} = \partial^2\Gamma$ *defines a canonical homeomorphism* $\hat{G}/\mathbb{R} \leftrightarrow \partial^2\Gamma$, *where the projection* $D\colon \hat{G} \longrightarrow \hat{G}/\mathbb{R} = \partial^2\Gamma$ *is a locally trivial fibration.* (Since $\partial^2\Gamma$ is paracompact and $\mathbb{R}$ is contractible this fibration is trivial, that is $\hat{G} \approx \partial^2\Gamma \times \mathbb{R}$. Yet there is no *canonical* homeomorphism between $\hat{G}$ and $\partial^2\Gamma$.) As it follows from the construction, the map $D\colon \hat{G} \longrightarrow \partial^2\Gamma$ is $\Gamma \times \mathbb{Z}_2$-equivariant for the canonical Γ action on $\partial^2\Gamma$ (coming from that on $\partial\Gamma$) and the involution $(a,b) \leftrightarrow (b,a)$ on $\partial^2\Gamma$.

(4) *If two* $\mathbb{R}$*-orbits* $g_1(t)$ *and* $g_2(t)$ *converge for* $t \longrightarrow \infty$, *that is if* $g_1(\infty) = g_2(\infty)$, *then the distance from* $g_1(t)$ *to* $\bar{g}_2 = \bigcup_t g_2(t) \subset \hat{G}$ *exponentially decays for* $t \longrightarrow \infty$. *That is* $d(t) = \mathrm{dist}(g_1(t), \bar{g}_2) \leqslant \text{const} \exp -\epsilon t$, *for some* const > 0 *and* $\epsilon > 0$. *Moreover, there exists an* $\epsilon = \epsilon(\hat{G}) > 0$, *such that the distance* $d(t)$ *for every two (convergent or not) orbits satisfies*

$$(++) \qquad d\left(\frac{t_1+t_2}{2}\right) \leqslant \exp(-\epsilon\,|t_1-t_2| + d(t_1) + d(t_2)),$$

for all t_1 *and* t_2 *in* $\mathbb{R}$.

(5) *Let* X *be an ultracomplete hyperbolic metric space which is isometrically acted upon by* Γ *such the action is* B-*finite and cobounded. Then there exists a (non-canonical) continuous quasi-isometric*

$\Gamma \times \mathbb{Z}_2$-equivarient map GX $\longrightarrow \hat{G}$ *which homeomorphically maps every geodesic in* GX *onto an* $\mathbb{R}$*-orbit in* $\hat{G}$.

(6) *The space* $\hat{G}$ *satisfying* (1), (2) *and* (3) *is unique in the following sense. For every two such, say* $\hat{G}_1$ *and* $\hat{G}_2$, *there exists a (non-canonical)* $\Gamma \times \mathbb{Z}_2$*-equivariant homeomorphism* $\hat{G}_1 \longleftrightarrow \hat{G}_2$ *which maps* $\mathbb{R}$*-orbits in* $\hat{G}_1$ *onto those in* $\hat{G}_2$.

<u>**Proof:**</u> Start with some X as in (5), take two points a and b on ∂X, consider two geodesics g_1 and g_2 in X between a and b and let us produce an equivalence relation between points $x_1 = g_1(t_1) \in g_1$ and $x_2 = g_2(t_2) \in g_2$ on these geodesics, such that the resulting quotient space will be our $\hat{G}$. First we fix a point $x_0 \in X$ and take a non-negative continuous function $\psi(x) = \psi(x,x_0)$ for $x \in X$ of the form $\psi(x,x_0) = \varphi(|x-x_0|)$, where $\varphi(\rho) = 1$ for $\rho \leqslant \rho_0$ and $\varphi(\rho) = 0$ for $\rho < \rho_1$ for two numbers $0 < \rho_0 < \rho_1 < \infty$. Next we take a non-negative continuous function $\eta = \eta(a,b)$ on $\partial^2 X$ of the form $\eta(a,b) = \zeta(|a-b|)$ for some metric on ∂X such that

(i) if every geodesic in X between a and b meets the ball in X of radius ρ_0' around x_0, then $\eta(a,b) = 1$;

(ii) if some geodesic in X between a and b misses the ball of radius ρ_1' around x_0, then $\eta(a,b) = 0$.

Here ρ_0' and ρ_1' are some constants in the interval $0 < \rho_0' < \rho_1' < \rho_0$.

Now take a geodesic $g = g(t)$ in X between two points a and b in ∂X such that $\eta(a,b) \neq 0$ and let

$$L_g(t) = \int_{t_0}^{t} \psi(g(t))dt,$$

where t_0 is the "center of gravity" of the function $\psi(g(t))$. That is

t_0 is determined by the equality

$$\int_{-\infty}^{+\infty} (t_0-t)\psi(g(t))dt = 0.$$

Then for points $x_1 = g(t_1)$ and $x_2 = g_2(t_2)$ on two different geodesics between a and b we set

$$L_0(a,b;x_1,x_2) = \eta(a,b)(L_{g_1}(t_1)-L_{g_2}(t_2))$$

where L_0 is extended by zero to those (a,b) where L_g were not previously defined.

Since the action of Γ is B-finite on X (and $\partial^3 X$) we can define

$$L(a,b;x_1,x_2) = \sum_{\gamma\in\Gamma} L_0(\gamma a,\gamma b,\gamma x_1,\gamma x_2).$$

8.3.D Let us enumerate basic properties of the function L.

(1) L is Γ-invariant.

(2) L is a cocycle in (x_1,x_2). That is $L(a,b;x_1,x_2) = -L(a,b;x_2,x_1)$ and $L(a,b;x_1,x_2) + L(a,b;x_2,x_3) + L(a,b;x_3,x_1) = 0$.

(3) $L(a,b;x_1,x_2) = -L(b,a;x_1,x_2)$.

(4) If ρ_0' is sufficiently large, then $|L(a,b;x_1,x_2)| > 0$ for all $a \neq b$ and all points $x_1 \neq x_2$ which lie on *same* geodesic between a and b. This follows from the coboundness of the action of Γ on X.

Now, assuming ρ_0' large, we identify any two geodesics between a and b, say g_1 and g_2 by the equivalence relation $x_1 \sim x_2 \Leftrightarrow L(a,b;x_1,x_2) = 0$ for $x_i \in g_i$, i = 1,2, and define $\hat{G}$ to be the resulting quotient space. If we start with a locally compact X, for example with the polyhedron $P_d^1(\Gamma)$, then $\hat{G}$ with the quotient topology is (obviously) locally compact and finite dimensional and (1),